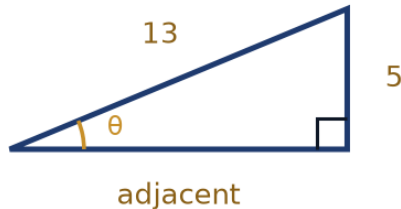


Trigonometry



Right-triangle trigonometric ratios

- 1 In a right triangle, angle θ has an opposite side of 5 and hypotenuse 13. Find $\sin \theta$ and the length of the adjacent side.



- 2 In a right triangle, angle θ has an opposite side of 15 and hypotenuse 17. Find $\cos \theta$.
- 3 In a right triangle, angle θ has an adjacent side of 7 and hypotenuse 25. Find $\sin \theta$ and $\tan \theta$.
- 4 In a right triangle, the two legs are 20 and 21, and the hypotenuse is 29. Find $\sin \theta$ and $\cos \theta$ for the angle θ opposite the side of length 20.

Special angles

- 5 State the exact value of $\sin(30^\circ)$.
- 6 State the exact value of $\cos(45^\circ)$.
- 7 State the exact value of $\tan(60^\circ)$.
- 8 State the exact values of $\sin(90^\circ)$ and $\cos(0^\circ)$.

Converting between radians and degrees

- 9 Convert 180° to radians.
- 10 Convert 90° to radians.
- 11 Convert $\frac{5\pi}{6}$ radians to degrees.
- 12 Convert 270° to radians.

The Pythagorean identity

- 13 Given $\sin \theta = \frac{3}{5}$ with θ acute, find $\cos \theta$ using $\sin^2 \theta + \cos^2 \theta = 1$.
- 14 Given $\cos \theta = \frac{5}{13}$ with θ acute, find $\sin \theta$.
- 15 Given $\sin \theta = 0.6$, find $\cos^2 \theta$.

Simplifying trigonometric expressions

- 16 Simplify $\sin^2 \theta + \cos^2 \theta + 1$.
- 17 Simplify $\frac{1 - \cos^2 \theta}{\sin \theta}$ (for $\sin \theta \neq 0$).
- 18 Simplify $\tan \theta \cos \theta$ (for $\cos \theta \neq 0$).

Graphs: amplitude and period

- 19 State the amplitude and period of $y = 5\sin(x)$.
- 20 State the amplitude and period of $y = 3\cos(2x)$.
- 21 State the amplitude and period of $y = 2\sin\left(\frac{x}{3}\right)$.
- 22 A sine graph has amplitude 4 and period π . Write its equation in the form $y = A\sin(Bx)$.

Graphs: phase shift and vertical shift

- 23 The graph of $y = \sin(x)$ is shifted $\frac{\pi}{3}$ to the right and 2 units up. Write the new equation.
- 24 State the vertical shift and midline of $y = 3\cos(x) + 5$.
- 25 The graph of $y = \cos(x)$ is shifted $\frac{\pi}{4}$ to the left. Write the new equation.

Solving basic trigonometric equations

- 26 Solve $2\sin \theta = 1$ for $0^\circ \leq \theta < 360^\circ$.

27 Solve $\sqrt{2}\cos\theta = 1$ for $0^\circ \leq \theta < 360^\circ$.

28 Solve $\tan\theta = -1$ for $0^\circ \leq \theta < 360^\circ$.

29 Solve $2\cos\theta + 1 = 0$ for $0^\circ \leq \theta < 360^\circ$.

Reference angles in other quadrants

30 Find $\sin(150^\circ)$ using the reference angle.

31 Find $\cos(240^\circ)$ using the reference angle.

32 Find $\tan(300^\circ)$ using the reference angle.

Complementary angle relationships

33 Given $\sin(20^\circ) \approx 0.342$, use the relationship $\sin\theta = \cos(90^\circ - \theta)$ to find $\cos(70^\circ)$.

34 Given $\sin(35^\circ) \approx 0.574$, find $\cos(55^\circ)$.

The unit circle: coordinates of key points

35 State the coordinates $(\cos\theta, \sin\theta)$ of the point on the unit circle at $\theta = 60^\circ$.

36 State the coordinates of the point on the unit circle at $\theta = 180^\circ$.

37 State the coordinates of the point on the unit circle at $\theta = 270^\circ$.

Word problems: angles of elevation

38 This question has two parts. A drone hovers directly above a landmark. From a point 200 meters from the base of the landmark, the angle of elevation to the drone is 35° .

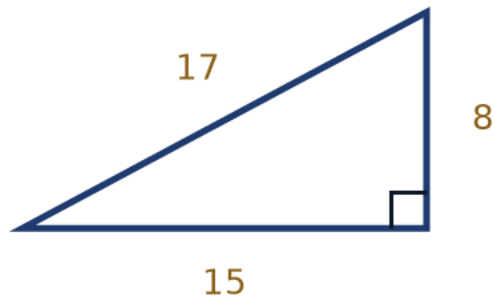
a Write an equation relating the drone's height h to the given information, using the tangent ratio.

b Find h , to the nearest meter, given $\tan(35^\circ) \approx 0.700$.

- 39 This question has two parts. A ladder 10 meters long leans against a wall, making an angle of 65° with the ground.
- a Write an equation for the height h the ladder reaches up the wall, using the sine ratio.
 - b Find h , to the nearest tenth of a meter, given $\sin(65^\circ) \approx 0.906$.
-

MCQ — Right-triangle trigonometric ratios

- 40 In the right triangle shown, angle θ has opposite side 8 and hypotenuse 17. Find $\sin \theta$.

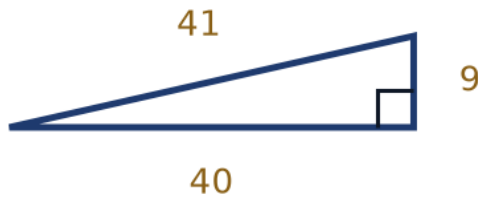


- a. $\frac{17}{8}$
 - b. $\frac{8}{15}$
 - c. $\frac{15}{17}$
 - d. $\frac{8}{17}$
- 41 In a right triangle, angle θ has opposite side 9 and hypotenuse 41. Find $\cos \theta$.
- a. $\frac{9}{41}$
 - b. $\frac{9}{40}$
 - c. $\frac{41}{40}$
 - d. $\frac{40}{41}$

42 In a right triangle, angle θ has adjacent side 20 and hypotenuse 29. Find $\sin \theta$.

- a. $\frac{21}{20}$
- b. $\frac{20}{29}$
- c. $\frac{21}{29}$
- d. $\frac{29}{21}$

43 In the right triangle shown, the two legs are 9 and 40, and the hypotenuse is 41. Find $\sin \theta$ and $\cos \theta$ (where θ is opposite the side of length 9).



- a. $\sin \theta = \frac{9}{40}, \cos \theta = \frac{40}{9}$
- b. $\sin \theta = \frac{40}{41}, \cos \theta = \frac{9}{41}$
- c. $\sin \theta = \frac{41}{9}, \cos \theta = \frac{41}{40}$
- d. $\sin \theta = \frac{9}{41}, \cos \theta = \frac{40}{41}$

44 In a right triangle, angle θ has opposite side 7 and adjacent side 24. Find $\tan \theta$.

- a. $\frac{24}{7}$
- b. $\frac{24}{25}$
- c. $\frac{7}{25}$
- d. $\frac{7}{24}$

MCQ — Special angles

45 State the exact value of $\cos(30^\circ)$.

a. 1

b. $\frac{\sqrt{3}}{2}$

c. $\frac{\sqrt{2}}{2}$

d. $\frac{1}{2}$

46 State the exact value of $\sin(45^\circ)$.

a. 1

b. $\frac{\sqrt{3}}{2}$

c. $\frac{\sqrt{2}}{2}$

d. $\frac{1}{2}$

47 State the exact value of $\tan(30^\circ)$.

a. $\frac{\sqrt{3}}{3}$

b. $\frac{1}{2}$

c. $\frac{\sqrt{3}}{2}$

d. $\sqrt{3}$

48 State the exact values of $\cos(90^\circ)$ and $\sin(0^\circ)$.

a. $\cos(90^\circ) = 0$, $\sin(0^\circ) = 1$

b. $\cos(90^\circ) = 1$, $\sin(0^\circ) = 0$

c. Both equal 1

d. Both equal 0

49 State the exact value of $\tan(45^\circ)$.

a. Undefined

b. $\sqrt{2}$

c. 1

d. 0

MCQ — Converting between radians and degrees

50 Convert 360° to radians.

a. 4π

b. π

c. 2π

d. $\frac{\pi}{2}$

51 Convert 45° to radians.

a. $\frac{\pi}{6}$

b. $\frac{\pi}{4}$

c. $\frac{\pi}{2}$

d. $\frac{\pi}{8}$

52 Convert $\frac{3\pi}{4}$ radians to degrees.

a. 120°

b. 150°

c. 135°

d. 270°

53 Convert 210° to radians.

- a. $\frac{7\pi}{12}$
- b. $\frac{7\pi}{3}$
- c. $\frac{7\pi}{6}$
- d. $\frac{5\pi}{6}$

MCQ — The Pythagorean identity

54 Given $\sin \theta = \frac{4}{5}$ with θ acute, find $\cos \theta$.

- a. $\frac{3}{5}$
- b. $\frac{4}{5}$
- c. $\frac{5}{3}$
- d. $\frac{9}{25}$

55 Given $\cos \theta = \frac{7}{25}$ with θ acute, find $\sin \theta$.

- a. $\frac{576}{625}$
- b. $\frac{24}{25}$
- c. $\frac{25}{24}$
- d. $\frac{7}{25}$

56 Given $\sin \theta = 0.8$, find $\cos^2 \theta$.

- a. 0.2
- b. 0.64
- c. 0.36
- d. 0.8

57 Given $\cos \theta = \frac{12}{13}$ with θ acute, find $\sin \theta$.

- a. $\frac{5}{13}$
- b. $\frac{13}{5}$
- c. $\frac{12}{13}$
- d. $\frac{25}{169}$

MCQ — Simplifying trigonometric expressions

58 Simplify $\sin^2 \theta + \cos^2 \theta - 1$.

- a. $2\sin^2 \theta$
- b. 0
- c. 2
- d. 1

59 Simplify $\frac{1 - \sin^2 \theta}{\cos \theta}$ (for $\cos \theta \neq 0$).

- a. $\frac{1}{\cos \theta}$
- b. $\cos \theta$
- c. $1 - \cos \theta$
- d. $\sin \theta$

60 Simplify $\frac{\sin \theta}{\tan \theta}$ (for $\tan \theta \neq 0$).

- a. 1
- b. $\sin \theta$
- c. $\cos \theta$
- d. $\frac{1}{\cos \theta}$

61 Simplify $2\sin^2 \theta + 2\cos^2 \theta$.

- a. 1
- b. 0
- c. 2
- d. $4\sin \theta \cos \theta$

MCQ — Graphs: amplitude and period

62 State the amplitude and period of $y = 4\sin(x)$.

- a. Amplitude 2π , period 4
- b. Amplitude 4, period 2π
- c. Amplitude 1, period 4
- d. Amplitude 4, period π

63 State the amplitude and period of $y = 2\cos(3x)$.

- a. Amplitude 2, period $\frac{2\pi}{3}$
- b. Amplitude 3, period $\frac{2\pi}{2}$
- c. Amplitude 6, period 2π
- d. Amplitude 2, period 6π

64 State the amplitude and period of $y = 5\sin\left(\frac{x}{2}\right)$.

- a. Amplitude 5, period π
- b. Amplitude 5, period 4π
- c. Amplitude $\frac{5}{2}$, period 2π
- d. Amplitude 2, period 4π

65 A cosine graph has amplitude 3 and period 4π . Write its equation in the form $y = A\cos(Bx)$.

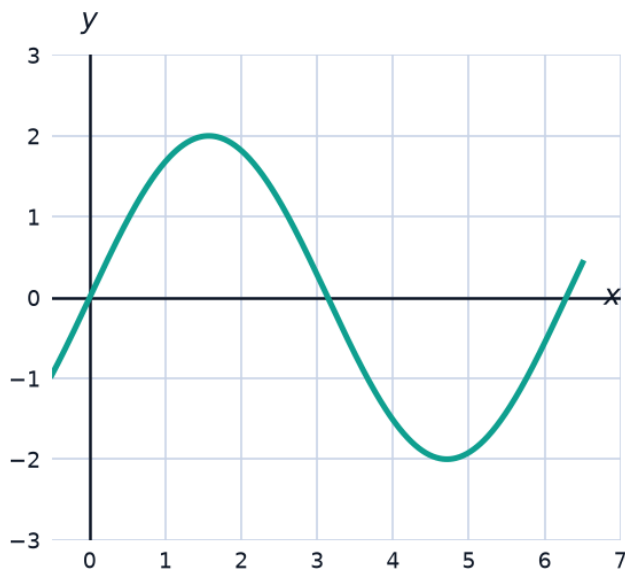
a. $y = 3\cos(4\pi x)$

b. $y = 4\pi\cos(3x)$

c. $y = 3\cos\left(\frac{x}{2}\right)$

d. $y = 3\cos(2x)$

66 The graph shows $y = A\sin(x)$ for some A . Find A by reading the graph's amplitude.



a. $A = 1$

b. $A = 2\pi$

c. $A = 4$

d. $A = 2$

MCQ — Graphs: phase shift and vertical shift

67 The graph of $y = \cos(x)$ is shifted $\frac{\pi}{4}$ to the right and 3 units up. Write the new equation.

a. $y = \cos(x) + \frac{\pi}{4} + 3$

b. $y = \cos\left(x - \frac{\pi}{4}\right) + 3$

c. $y = \cos\left(x + \frac{\pi}{4}\right) + 3$

d. $y = \cos\left(x - \frac{\pi}{4}\right) - 3$

68 State the vertical shift and midline of $y = 2\sin(x) - 4$.

a. Down 4, midline $y = 2$

b. Up 4, midline $y = 4$

c. Down 4, midline $y = -4$

d. Down 2, midline $y = -4$

69 The graph of $y = \sin(x)$ is shifted $\frac{\pi}{6}$ to the left. Write the new equation.

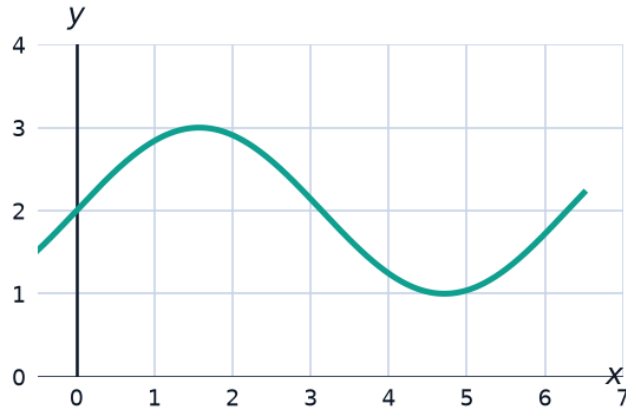
a. $y = \sin\left(\frac{\pi}{6} - x\right)$

b. $y = \sin(x) + \frac{\pi}{6}$

c. $y = \sin\left(x + \frac{\pi}{6}\right)$

d. $y = \sin\left(x - \frac{\pi}{6}\right)$

- 70 The graph shows $y = \sin(x)$ shifted vertically. Find the new midline by reading the graph.



- a. $y = 0$
- b. $y = 2$
- c. $y = 1$
- d. $y = -2$

MCQ — Solving basic trigonometric equations

- 71 Solve $2\sin \theta = \sqrt{2}$ for $0^\circ \leq \theta < 360^\circ$.

- a. $\theta = 45^\circ$ only
- b. $\theta = 45^\circ$ or $\theta = 135^\circ$
- c. $\theta = 45^\circ$ or $\theta = 225^\circ$
- d. $\theta = 60^\circ$ or $\theta = 120^\circ$

- 72 Solve $\sqrt{3}\tan \theta = 3$ for $0^\circ \leq \theta < 360^\circ$.

- a. $\theta = 60^\circ$ or $\theta = 300^\circ$
- b. $\theta = 30^\circ$ or $\theta = 210^\circ$
- c. $\theta = 60^\circ$ or $\theta = 240^\circ$
- d. $\theta = 60^\circ$ only

73 Solve $2\cos \theta = 1$ for $0^\circ \leq \theta < 360^\circ$.

a. $\theta = 30^\circ$ or $\theta = 330^\circ$

b. $\theta = 60^\circ$ or $\theta = 240^\circ$

c. $\theta = 60^\circ$ only

d. $\theta = 60^\circ$ or $\theta = 300^\circ$

74 Solve $\tan \theta = 1$ for $0^\circ \leq \theta < 360^\circ$.

a. $\theta = 45^\circ$ or $\theta = 225^\circ$

b. $\theta = 45^\circ$ only

c. $\theta = 45^\circ$ or $\theta = 315^\circ$

d. $\theta = 45^\circ$ or $\theta = 135^\circ$

75 Solve $2\sin \theta + 1 = 0$ for $0^\circ \leq \theta < 360^\circ$.

a. $\theta = 210^\circ$ or $\theta = 300^\circ$

b. $\theta = 150^\circ$ or $\theta = 330^\circ$

c. $\theta = 30^\circ$ or $\theta = 150^\circ$

d. $\theta = 210^\circ$ or $\theta = 330^\circ$

MCQ — Reference angles in other quadrants

76 Find $\sin(120^\circ)$ using the reference angle.

a. $\frac{\sqrt{3}}{2}$

b. $-\frac{\sqrt{3}}{2}$

c. $\frac{1}{2}$

d. $-\frac{1}{2}$

77 Find $\cos(210^\circ)$ using the reference angle.

- a. $-\frac{\sqrt{3}}{2}$
- b. $\frac{\sqrt{3}}{2}$
- c. $-\frac{1}{2}$
- d. $\frac{1}{2}$

78 Find $\tan(330^\circ)$ using the reference angle.

- a. $-\frac{\sqrt{3}}{3}$
- b. $\sqrt{3}$
- c. $-\sqrt{3}$
- d. $\frac{\sqrt{3}}{3}$

79 Find $\sin(315^\circ)$ using the reference angle.

- a. $-\frac{\sqrt{2}}{2}$
- b. $-\frac{1}{2}$
- c. $\frac{\sqrt{2}}{2}$
- d. $\frac{1}{2}$

MCQ — Complementary angle relationships

80 Given $\sin(25^\circ) \approx 0.423$, use the relationship $\sin \theta = \cos(90^\circ - \theta)$ to find $\cos(65^\circ)$.

- a. -0.423
- b. 0.906
- c. 0.577
- d. 0.423

81 Given $\cos(40^\circ) \approx 0.766$, find $\sin(50^\circ)$.

- a. 0.643
- b. 0.766
- c. 1.306
- d. 0.234

82 Given $\sin(70^\circ) \approx 0.940$, find $\cos(20^\circ)$.

- a. 1.064
- b. 0.060
- c. 0.342
- d. 0.940

MCQ — The unit circle: coordinates of key points

83 State the coordinates $(\cos \theta, \sin \theta)$ of the point on the unit circle at $\theta = 30^\circ$.

- a. $\left(\frac{1}{2}, \frac{\sqrt{3}}{2}\right)$
- b. $\left(\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2}\right)$
- c. $(1, 0)$
- d. $\left(\frac{\sqrt{3}}{2}, \frac{1}{2}\right)$

84 State the coordinates of the point on the unit circle at $\theta = 90^\circ$.

- a. $(1, 0)$
- b. $(0, -1)$
- c. $(0, 1)$
- d. $(-1, 0)$

85 State the coordinates of the point on the unit circle at $\theta = 225^\circ$.

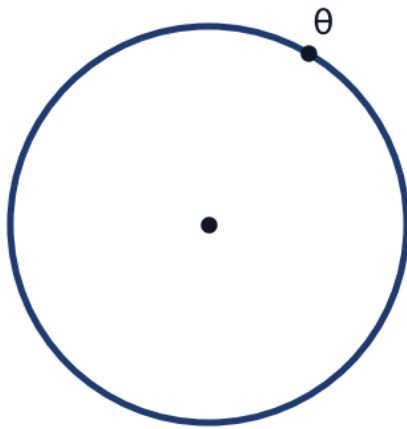
a. $\left(\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2}\right)$

b. $\left(-\frac{\sqrt{2}}{2}, -\frac{\sqrt{2}}{2}\right)$

c. $(-1, -1)$

d. $\left(-\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2}\right)$

86 The unit circle shown marks a point at angle $\theta = 60^\circ$. What are its coordinates?



a. $\left(\frac{\sqrt{3}}{2}, 0.5\right)$

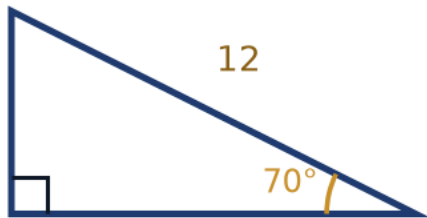
b. $(0.5, 0.5)$

c. $(1, 0)$

d. $\left(0.5, \frac{\sqrt{3}}{2}\right)$

MCQ — Word problems: angles of elevation

- 87 A drone hovers directly above a landmark. From a point 150 meters from the landmark's base, the angle of elevation to the drone is 40° . Find the drone's height, to the nearest meter.
- a. 150
 - b. 196
 - c. 126
 - d. 115
- 88 A ladder leans against a wall, making an angle of 70° with the ground. If the ladder shown is 12 meters long, find how high up the wall it reaches, to the nearest meter.



- a. 4
 - b. 12
 - c. 8
 - d. 11
- 89 From the top of an 80 m cliff, the angle of depression to a boat is 25° . Find the horizontal distance from the base of the cliff to the boat, to the nearest meter.
- a. 37
 - b. 80
 - c. 172
 - d. 189